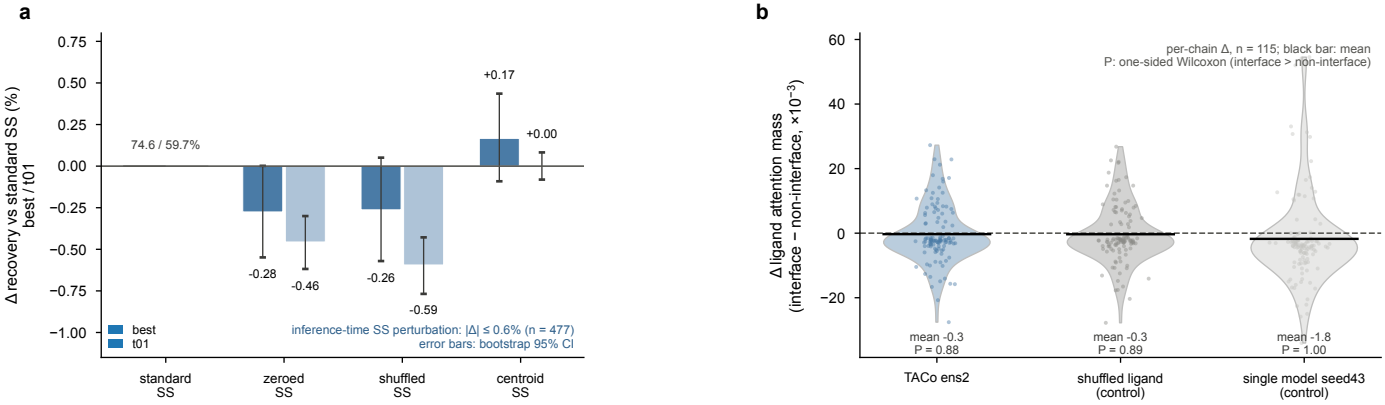
a
Δ recovery vs standard SS (%)
best / t01
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
74.6 / 59.7%
-0.28
-0.46
-0.26
-0.59
+0.17
+0.00
best
t01
inference-time SS perturbation: |Δ| ≤ 0.6% (n = 477)
error bars: bootstrap 95% CI
standard SS
zeroed SS
shuffled SS
centroid SS
b
per-chain Δ, n = 115; black bar: mean
P: one-sided Wilcoxon (interface > non-interface)
Δ ligand attention mass
(interface − non-interface, ×10⁻³)
60
40
20
0
−20
mean -0.3
P = 0.88
mean -0.3
P = 0.89
mean -1.8
P = 1.00
TACo ens2
shuffled ligand (control)
single model seed43 (control)